LexiSync: evidence-grounded legal document analyst

Every claim cites page + clause, is verified against its source, and the system abstains when the documents lack evidence.

INDEXING (once per document)

Documents
2–5 public legal PDFs
GDPR (88 p.), ePrivacy (11 p., two-column), any upload / URL
cases = separate document sets

→

1 Ingestion
PyMuPDF dict · NFKC · reading order
2-column bands · header/footer removal
footnotes · tables (find_tables)
scanned → Tesseract OCR 200 dpi

→

2 Clause-aware chunking
EU: Article 83 → 83(5), recitals, 4(11)
contract: 7. Liability → 7.2
merge < 120 chars · split > 1500
label 'GDPR p. 83, Article 83(5)' + bboxes

→

3 Index
BM25 (legal-ref tokenizer: 7.2, 83(5))
bge-small-en-v1.5 embeddings (384-d)
fingerprinted pickle cache
GDPR + ePrivacy: 747 chunks

index used by retrieval

ANSWERING (per question)

4 Query planner
splits 'A, and which B?'
vague part inherits topic
each part searched separately
best passage of each part kept

→

5 Hybrid retrieval
BM25 rank + dense rank
Reciprocal Rank Fusion, k = 60
top 30 candidates
(ranks, not raw scores)

→

6 Cross-encoder rerank
ms-marco-MiniLM-L-6-v2
reads query + passage together
sigmoid → relevance 0..1
top k = 4–6 passages

→

7 Retrieval gate
best relevance < 0.10 ?
→ ABSTAIN, no LLM call
list documents searched
(answerable ≥ 0.3, unanswerable ≈ 0)

relevance ≥ 0.10

8a LLM: qwen2.5:7b (Ollama)
temperature 0 · seed 42 · JSON schema
claims (text, [C#], quote) → answer
→ missing → abstain (decide last)
rules: only passages, no swapped numbers

←

8b Verifier (code)
quote in cited passage: rapidfuzz
partial_ratio ≥ 85 or overlap ≥ 0.6
every number must match (20M ≠ 10M)
fail → reformulate, retry (max 2)

failed claim → new query

←

9 Citations & confidence
[C2] → 'GDPR p. 83, Article 83(5)'
only cited, checked sentences shown
HIGH / MEDIUM / LOW / INSUFFICIENT EVIDENCE

←

10 UI (Streamlit)
answer + confidence badge
citation chips · timings
claim verification table
cited PDF page, region highlighted

Abstention: gate (step 7), model says no answer, or no claim survives the verifier → INSUFFICIENT EVIDENCE + list of documents searched

EVALUATION

20 questions (11 factual, 5 multi-hop incl. 3 cross-document, 4 unanswerable) + held-out set on an unseen document · recall@1/3/5/10 + MRR per retrieval mode

recall@3: BM25 0.61 · dense 0.77 · hybrid 0.71 · +rerank 0.83 · +planner 0.93 (MRR 0.93) · OCR CER 3.2 %

end-to-end correct: closed-book 35 % · dense-only 60 % · full 75 % → 80 % with second look (cit. precision 83 %) · unanswerable abstained 4/4 · held-out Basic Law: 7/8 correct, 3/3 refused